

Financial Time Series

46-929

Solutions #3

1. Download data on three-month treasury bill rates from the St. Louis Federal Reserve for 1/1/85 to 12/31/2005. You can download the series from:
research.stlouisfed.org/fred2/series/TB3MS?cid=116
 - a) Plot the data and indicate whether the data series seems to be stationary.
 - b) If the series is not apparently stationary, difference the series as few times as needed to reject the presence of a unit root using the Dickey Fuller test.
 - c) Fit an ARIMA(p,d,q) model to the series using the order_selection command, keeping d at the value you chose in the previous part. If the original series was considered to be stationary, then set $d = 0$.
 - d) Run diagnostic tests on the model. Does it seem to be a reasonable fit?

SOLUTION

- a) The following Python script will give answers to the questions below. Clearly the series is not stationary.

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np

from statsmodels.tsa.stattools import adfuller
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf
from statsmodels.tsa.stattools import arma_order_select_ic
from statsmodels.tsa.arima_model import ARIMA

us_tb = pd.read_csv('TB3MS.csv')
us_tb = us_tb.iloc[(1985-1934)*12:-(2019-2005)*12-3]

adfuller(us_tb['TB3MS'].values)

adfuller(np.diff(us_tb['TB3MS'].values, 1))
plot_acf(np.diff(us_tb['TB3MS'].values, 1))
plt.show()
plot_pacf(np.diff(us_tb['TB3MS'].values, 1))
plt.show()
arma_order_select_ic(np.diff(us_tb['TB3MS'].values,1),ic=['aic', 'bic'])
model = ARMA(np.diff(us_tb['TB3MS'].values, (3,0)).fit(dis=0))
model = ARIMA(us_tb['TB3MS'].values, (1, 1, 0)).fit(dis=0)
```

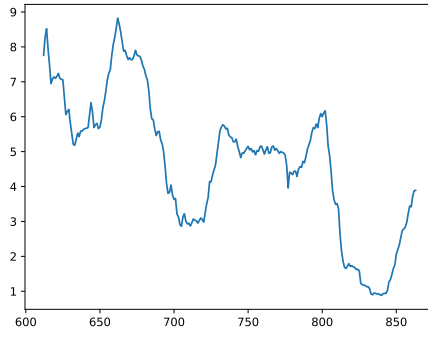
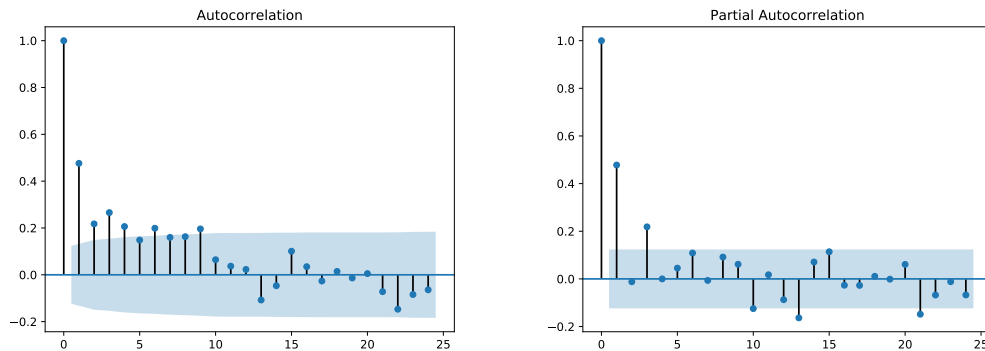


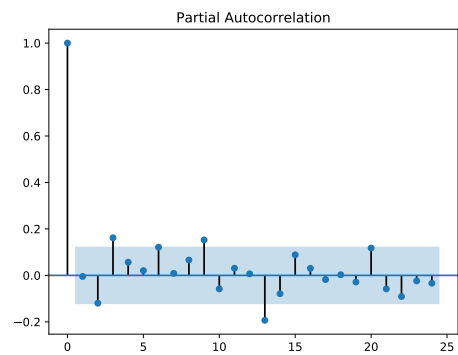
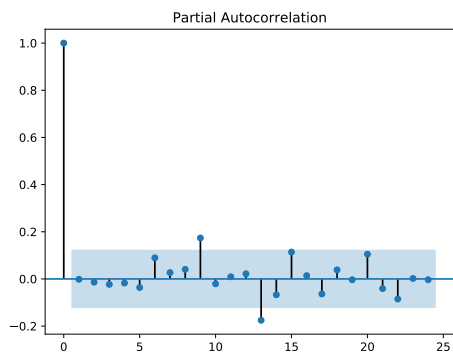
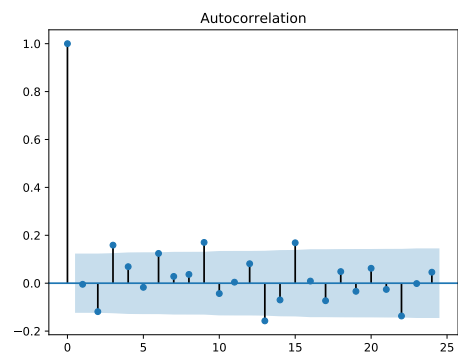
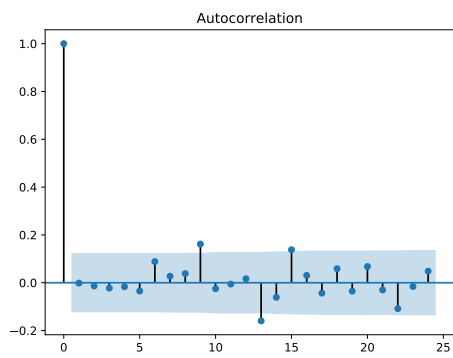
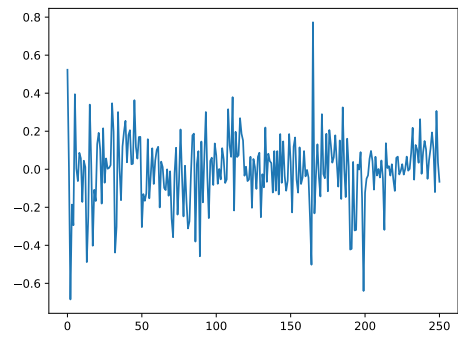
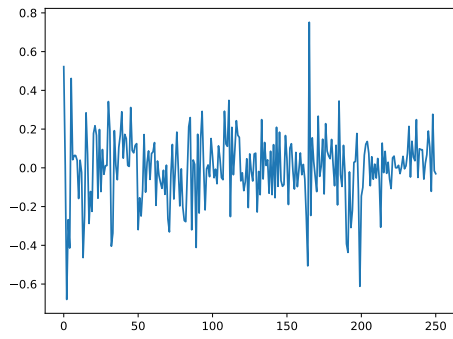
Figure 1: Raw data

- b) The Dickey Fuller test on the raw data does not reject the null hypothesis of a unit root as the p-value is .146: $(-2.3835424188318792, 0.1464354168524229, 10, 241)$. When the data is differenced one time, we get a p-value of $1.04e-06$, and reject the null hypothesis. $(-5.639612366776882, 1.0437937963045998e-06, 2, 248)$. We now conduct analysis with the differenced data.
- c) We next show the ACF and PACF for the differenced sequence. The PACF shows that we need to include at least 3 lags for an AR model.



The ARMA order select command gives is (3,0) for AIC and (1,0) for BIC, which means ARIMA(3,1,0) or ARIMA(1,1,0) for the original data set. The BIC result is suspicious since the PACF above indicated we need more lags than 1.

- d) We show the residuals, the ACF and the PACF figures side-by-side for the ARIMA(3,1,0) on the left and ARIMA(1,1,0) on the right. From the ACF and PACF, it is clear that the ARIMA(3,1,0) model is the best.

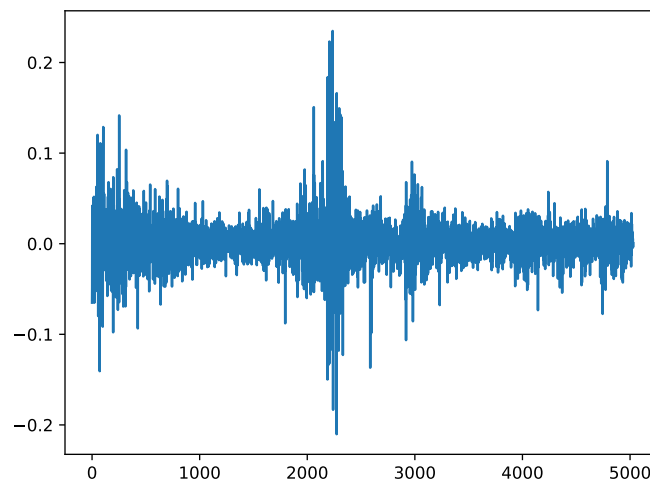


2. Consider any stock of your own choosing and any time period of at least 10 years length.
 - a) Plot the one-day log returns using the Adjusted Close data.
 - b) Using the adjusted Dickey Fuller test, assess whether the series is weakly stationary. If not, can you make it stationary by differencing?
 - c) Plot the ACF and PACF. What do you see?
 - d) Use the Arma order_select command in Statsmodels. Using the aic and bic criteria, find the suggested appropriate ARMA model(s),
 - e) From the models selected and other similar models, present the results from the best model in your judgment. Does it provide a good fit to these data? Does it pass the Ljung-Box and the adjusted Dickey Fuller tests?

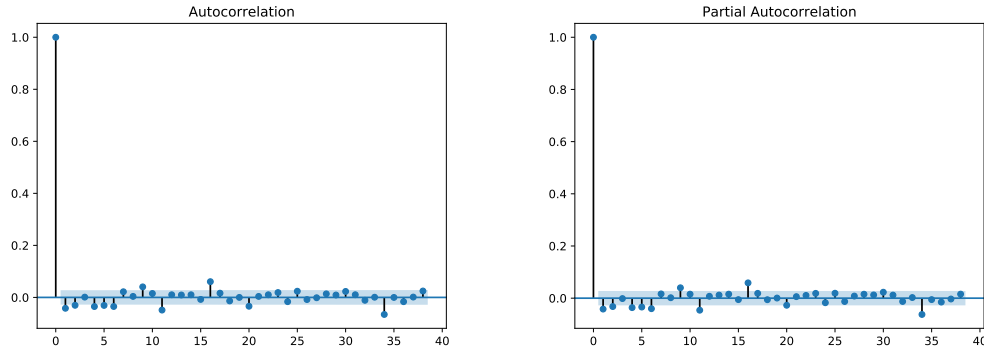
Solution

Using Goldman-Sachs for 1/1/2000 to 12/31/2019.

- a) Plot the one-day log returns using the Adjusted Close data.



- b) Using the adjusted Dickey Fuller test, assess whether the series is weakly stationary. If not, can you make it stationary by differencing?
 The adjusted Dickey-Fuller test clearly rejects the null hypothesis of a unit root with tiny p-value of 1.26e-29.
 (-16.782971758161047, 1.2598274349939462e-29, 15, 5014,)
- c) Plot the ACF and PACF. What do you see?



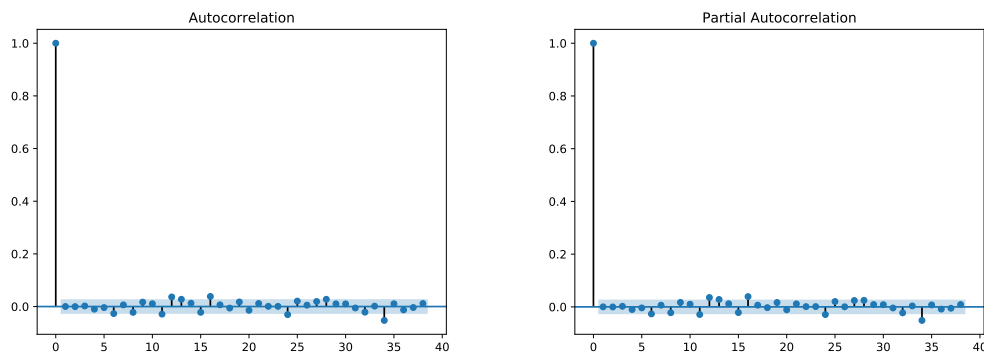
- d) Use the `Arma_order_select` command in `Statsmodels` to find the suggested appropriate ARMA model(s),

```
res = sm.tsa.arma_order_select_ic(logret_ibm_shf, ic=['aic', 'bic'], trend='nc')
print(res.aic_min_order) (4,2)
print(res.bic_min_order) (1,1)
```

- e) From the models selected, present the results from the best model. Does it provide a good fit to these data?

The two selected models based on the AIC and BIC provide very good fits to the data, at least from looking at the ACF and PACF plots of the residuals for both. All of the coefficients are essentially non-significant (I didn't print out the actual confidence interval endpoints), and are in most cases tiny. We could pick either, and visually (4,2) is a bit better than (1,1); however, at the cost of 6 parameters versus 2. It would seem that an ARMA(1,1) model provides a very good fit to the data, and it has been selected by the BIC criterion. Nevertheless, there are clear signs of stochastic volatility, so we could revisit this example when we have learned about GARCH models.

(4,2) model



ACF

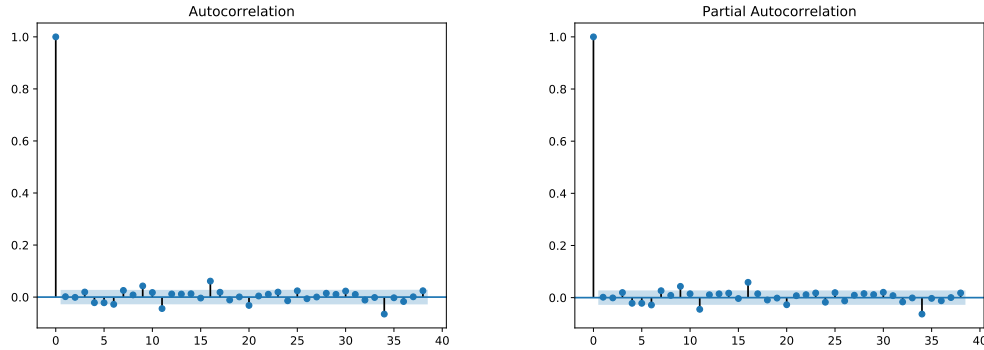
```
(array([ 1.00000000e+00, 1.79461280e-04, -2.44975335e-04, 2.17206131e-03, -1.00356145e-02,
        -3.85632875e-03, -2.66587744e-02, 6.33609644e-03, -2.20378382e-02, 1.68499359e-02,
        1.06757752e-02]), array([[ 1.00000000e+00, 1.00000000e+00]]),
```

PACF

```
(array([ 1.00000000e+00, 1.79496965e-04, -2.45105006e-04, 2.17344574e-03, -1.00444915e-02,
        -3.85567065e-03, -2.67021028e-02, 6.39801883e-03, -2.21982616e-02, 1.69631657e-02,
```

1.00800419e-02]), array([[1.00000000e+00, 1.00000000e+00],

(1,1) model



ACF

(array([1. , 0.00159288, -0.00102386, 0.01947379, -0.02160683, -0.02179755, -0.02776337, 0.02558894, 0.00840609, 0.04280169, 0.01761185]), array([[1.00000000e+00, 1.00000000e+00],

PACF

(array([1. , 0.00159319, -0.00102681, 0.01948875, -0.02169594, -0.0217181 , -0.02818144, 0.02655826, 0.00870158, 0.04319192, 0.01487181]), array([[1.00000000e+00, 1.00000000e+00],

3. (Ruppert and Matteson, problem 16, page 359)

Suppose you fit an AR(2) model to a time series Y_t , $1 \leq t \leq n$ and the estimates were $\hat{\mu} = 100.1$, $\hat{\phi}_1 = 0.5$ and $\hat{\phi}_2 = 0.1$. The last three observations were $Y_{n-2} = 101.0$, $Y_{n-1} = 99.5$, and $Y_n = 102.3$. What are the forecasts of Y_{n+1} , Y_{n+2} and Y_{n+3} ?

Solution

Using the AR(2) model $Y_n = \mu + \phi_1(Y_{n-1} - \mu) + \phi_2(Y_{n-2} - \mu) + \epsilon_n$, plugging in the estimates for μ, ϕ_1, ϕ_2 and estimating the future ϵ terms to be 0, we find

$$\hat{Y}_{n+1} = .4 \cdot \mu + .5Y_n + .1Y_{n-1} = 40.04 + .5 \cdot 102.3 + .1 \cdot 102.3 = 101.14$$

$$\hat{Y}_{n+2} = .4 \cdot \mu + .5\hat{Y}_{n+1} + .1Y_n = 40.04 + .5 \cdot 101.14 + .1 \cdot 102.3 = 100.84$$

$$\hat{Y}_{n+3} = .4 \cdot \mu + .5Y_{n+2} + .1Y_{n+1} = 40.04 + .5 \cdot 100.84 + .1 \cdot 101.14 = 100.574$$

4. (Ruppert and Matteson, problem 12, page 357)

The MA(2) model $Y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$ was fit to data and the estimates are $\hat{\mu} = 45$, $\hat{\theta}_1 = 0.3$, and $\hat{\theta}_2 = -0.15$. Suppose the last two values of the observed time series and residuals are $Y_{n-1} = 39.8$, $Y_n = 42.7$, $\hat{\epsilon}_{n-1} = -4.3$ and $\hat{\epsilon}_n = 1.5$. Find the forecasts of Y_{n+1} and Y_{n+2} .

Solution

$$Y_t = \mu + \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}.$$

$$\hat{Y}_{t+1} = \hat{\mu} + \hat{\epsilon}_{t+1} + \hat{\theta}_1\epsilon_t + \hat{\theta}_2\epsilon_{t-1} = 45 + 0 + .3(1.5) + (-.15)(-4.3) = 46.095$$

$$\hat{Y}_{t+2} = \hat{\mu} + \hat{\epsilon}_{t+2} + \hat{\theta}_1\hat{\epsilon}_{t+1} + \hat{\theta}_2\epsilon_t = 45 + 0 + 0 + (-.15)(1.5) = 44.775$$